MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL

Department of Electrical Engineering

B. Tech. IV Semester

Sub: Electronics Devices & Circuits (EE-223)

Assignment - 1

Submission deadline: 27.03.2025

Q.1. A voltage amplifier of the type of Fig. 1.1 is fed by a 200-mV source and drives a 10 Ω load. Voltmeter measurements give v_I = 150 mV and v_O = 10 V. If disconnecting the load causes v_O to rise to 12 V, and then connecting a 30-kΩ resistor across the input-port terminals causes v_O to drop from 12 V to 9.6 V, find R_s , R_i , A_{oc}, and R_o.

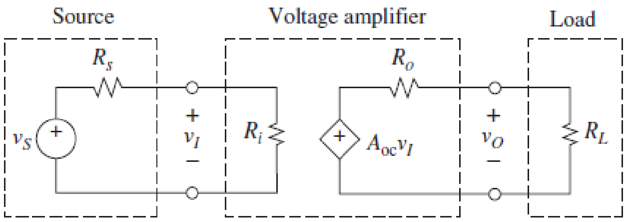

Fig. 1

Q.2. In Ltspice simulation platform, obtain the Input Offset Current, Input bias current, Input & Output Offset Voltage the Bandwidth, Slew Rate & Output Impedance of an operational Amplifier OP191.

Q.3. Design a noninverting amplifier whose gain is variable over the range 0.5 V/V $\leq A \leq$ 5 V/V by means of a 100-kΩ pot. Validate the designed non-inverting amplifier in Ltspice simulation platform.

Q.4. A source v_S = 2 V with R_s = 10 kΩ is to drive a gain-of-five inverting amplifier implemented with R_1 = 20 kΩ and R_f = 100 kΩ. Find the amplifier output voltage and verify that because of loading its magnitude is *less* than 2×5 = 10 V. (*b*) Find the value to which R_f must be changed if we want to compensate for loading and obtain a full output magnitude of 10 V. Validate the result in Ltspice simulation platform.

Q.5. Find v_N , v_P , and v_O in the circuit of Fig. 2, as well as the power released by the 4-V source. Validate the result in Ltspice simulation platform.

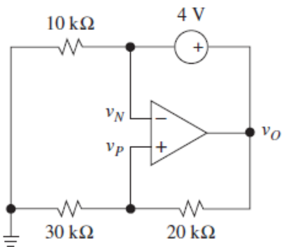

Fig. 2

Q.6. The audio panpot circuit of Fig. 3 is used to continuously vary the position of signal v_I between the left and the right stereo channels. (*a*) Discuss circuit operation. (*b*) Design R_1 and R_2 so that $v_L/v_I = -1$ V/V when the wiper is fully down, $v_R/v_I = -1$ V/V when the wiper is fully up, and $v_L/v_I = v_R/v_I = -1/\sqrt{2}$ when the wiper is halfway. Validate the design in Ltspice simulation platform.

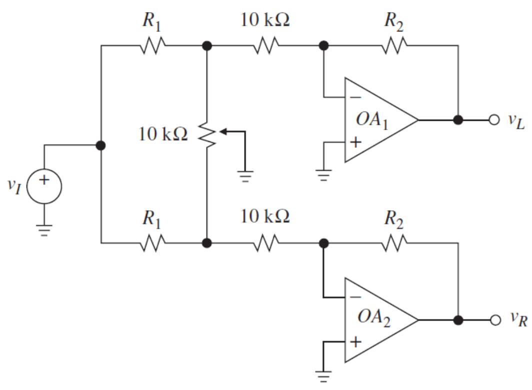

Fig. 3

Q.7. In the differentiator circuit based on Op-amp, let C = 10 nF and R = 100 kΩ, and let v_I be a periodic signal alternating between -2 V and 2 V with a frequency of 100kHz. Simulate and plot v_I and v_O versus time, if v_I is (*a*) a sine wave; (*b*) a triangular wave.

Q.8. In the integrator of Fig. 1.20 let R =100 k_ and C =10 nF. Simulate and plot $v_I(t)$ and $v_O(t)$ if $v_I = 5\sin(2\pi 100t)$V.

Q.9. An OP AMP circuit is shown in the Fig. 4 below:

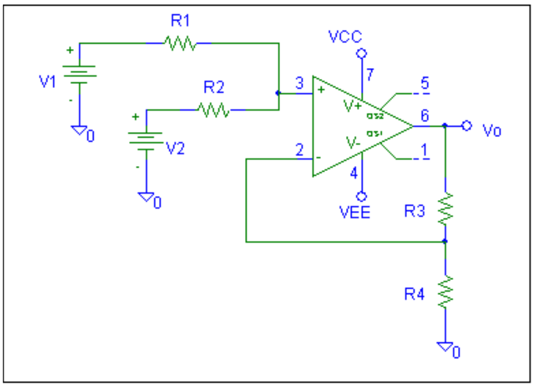

Fig. 4

Prove that the output is

$$Vo = \frac{1}{2} \times \left(1 + \frac{R_3}{R_4}\right) \times (V_1 + V_2)$$

If $R_1 = R_2 = R$, and input current drawn by the OP AMP is negligible.

Q.10. If charge on the capacitor is reset to zero, prove that the output voltage ***'V$_o$'*** in the circuit given in Fig. 5 below, is expressed by $V_O(t) = -\frac{1}{RC}\int_0^t (V_1 + V_2)$

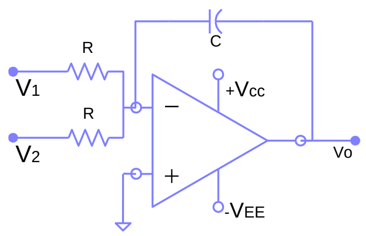

Fig. 5

Q.11. Find the frequency for which the amplifier gain V_O/V_S in Fig. 6 is minimum. What is the value of the minimum gain? Assume the op-amp to be ideal. Take $R=1k\Omega$ and $C=\pi\mu F$.

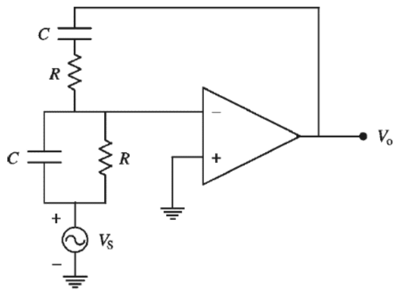

Fig. 6

Q.12. Full scale deflection current for dc microammeter used in the Fig. 7 is 50µA and its internal resistance is 2kΩ. A dc voltmeter of 0-10V is to be constructed using ammeter circuit shown. Find the value of R1. Which of the two terminals, + or -, of the microammeter is required to be grounded?

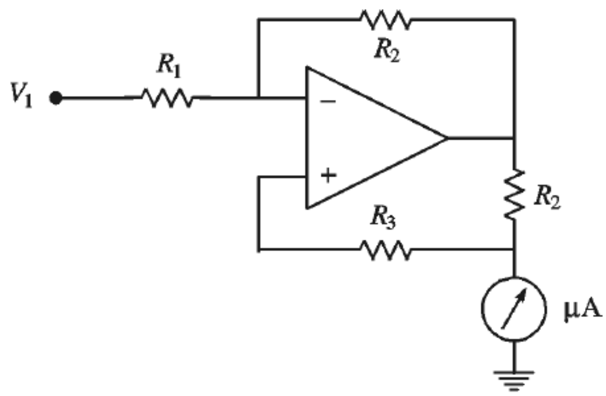

Fig. 7

Q.13. Using a 555 timer, design an astable multivibrator of frequency 250Hz and duty cycle „ D0"% (Where D is the last digit of your ID number. For example if your number is 20XXXXXX012H, the duty cycle should be 20%. If your ID number ends with a zero, the duty cycle has to be 25%).

Q.14. Prove that the circuit given in Fig. 8 below converts change in transducer resistance to voltage where offset voltage is zeroed by potentiometer

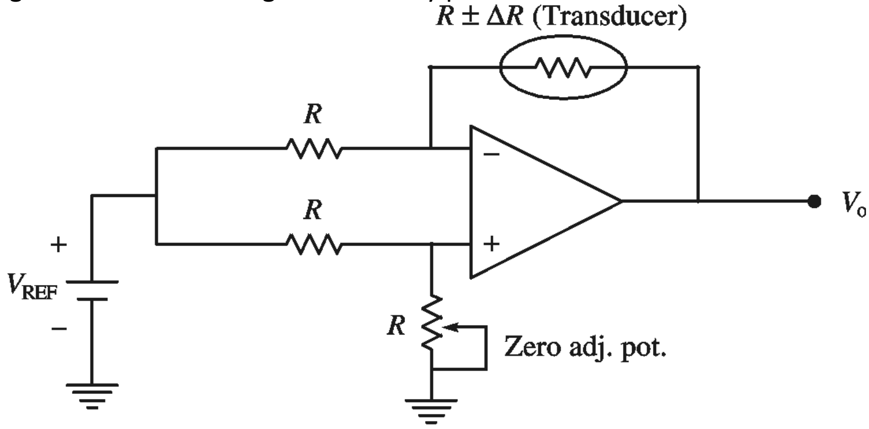

Fig. 8

Q.15. Find out the output voltage in the Fig. 9 below. Given that $+V=5V$, $V_Z = 3.3V$, $R=220\Omega$. For Fig.(a), the value of $R_F = 20k\ \Omega$, and $R_I = 10k\ \Omega$, and for Fig.(b) $R_F = R_I = 10k\ \Omega$

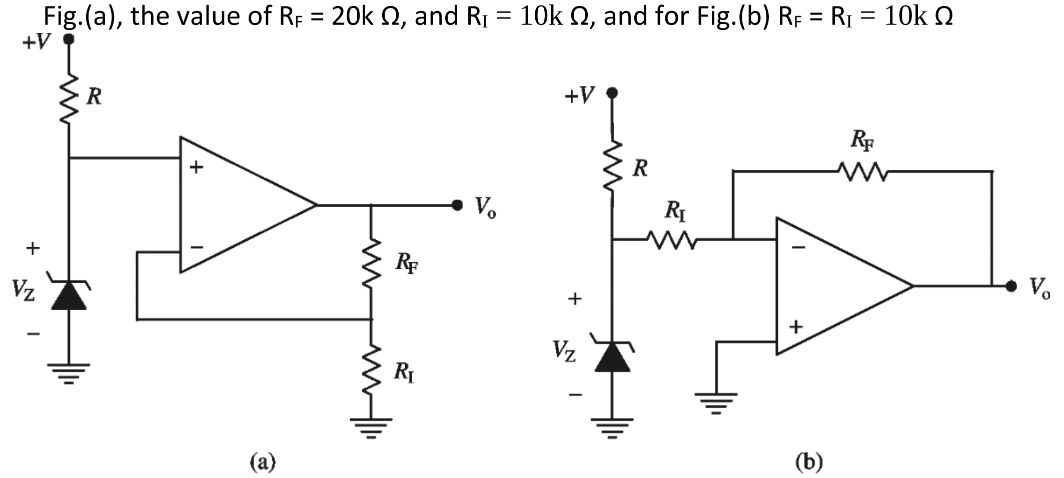

Fig. 9